

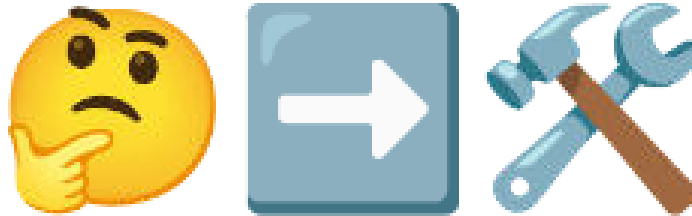
What Problem Can We Solve?

Engineering Design — Unit 1, Lesson 1 of 6

 **Grade:** 1st Grade

 **Duration:** 30–40 minutes

 **Subject:** Engineering Design



Engineers start by noticing a problem and imagining a way to help.

NGSS Standards:

K-2-ETS1-1 — Ask questions, make observations, and gather information about a situation people want to change to define a simple problem that can be solved through the development of a new or improved object or tool.

Learning Objective

Students will **identify** a simple everyday problem and **describe** who needs help and what a successful solution should do.

The Big Idea

Engineering begins with a problem. A problem is something that is hard to do or something we want to make better. Engineers look carefully, ask questions, and think about how to help people.

 **Key vocabulary:** problem, solution, engineer, improve, tool, help

Materials

- Pencil
- Crayons
- Worksheet pages from this lesson
- Optional: classroom or household objects to discuss



Let's Get Started

Step 1 — Notice Problems (5 min)

Talk about everyday examples: carrying many books, keeping papers from blowing away, reaching something high, keeping lunch cold. Ask your child which ones sound like real problems.

Step 2 — Who Needs Help? (8 min)

Pick one simple problem together. Ask: "Who has this problem?" and "What is hard about it?" Help the child explain the situation in plain words.

Step 3 — What Would Help? (8 min)

Ask: "What should a helpful tool or object do?" Focus on the job of the solution, not the fancy details. For example: *"It should help carry the books without dropping them."*

Step 4 — Draw the Problem (8 min)

Have the child draw the problem and circle the part that needs to change or improve.



Discussion Questions



Question 1: "What makes something a problem?"



A problem is something that is hard to do, something unsafe, or something we want to make better.



Question 2: "Who are we trying to help?"



Engineers help people. It is important to know who the user is.



Question 3: "What should a good solution do?"



*It should help with the problem. It does not need to be perfect at first, but it should do its job. This is the design **success rule**—what the tool must do to be helpful.*



Parent/Teacher Notes

- Keep problems simple and familiar.
- Do not rush into building. This lesson is about defining the problem clearly.
- Strong student response: "The problem is ____ because _____. A solution should help by _____."



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Student Worksheet — What Problem Can We Solve?

Name: _____ Date: _____

1. What is one problem you want to solve?

2. Who has this problem?

3. What should a good solution do?

4. Draw the problem:

Draw here 



PARENT/TEACHER ANSWER KEY — Detach before giving worksheet to students

Answers will vary. Accept any realistic everyday problem with a clear person/user and a sensible description of what a solution should do.

Sample response: *"The problem is carrying too many books. The user is a student. A helpful solution should hold the books together and stop them from falling."*



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Look at Animal Solutions



Engineering Design — Unit 1, Lesson 2 of 6



Grade: 1st Grade



Duration: 35–45 minutes



Subject: Engineering Design



Kid-friendly illustration of a duck paddling with webbed feet, a turtle with a strong shell, a bird using its beak, and a hedgehog showing protective spines.

Animals already solve real problems with their body parts. Engineers can borrow those ideas.



NGSS Standard:

1-LS1-1 — Use materials to design a solution to a human problem by mimicking how plants and/or animals use their external parts to help them survive, grow, and meet their needs.



Learning Objective

Students will **notice** how animal body parts do important jobs, **match** each structure to the problem it solves, and **choose** one idea a human tool could copy.



The Big Idea

Animals do not have tools from a store, but they still solve problems every day. A duck's webbed feet help it move through water. A turtle's shell helps protect its body. A bird's beak helps it pick up food. Engineers look at helpful structures like these and ask, "*Could we copy that idea to help a person?*"



Key vocabulary: structure, mimic, protect, scoop, grab, glide, solve



Success rule: By the end of the lesson, your child should be able to say: "*This animal part helps with _____. A human tool could copy it to help with _____.*"



The Science + Engineering Connection

In this unit, children are not just naming animal parts. They are learning to think like little engineers. First they look closely at nature. Then they figure out what job a body part does. Then they borrow the useful idea. That is called **mimicking** nature.

For first grade, the important thinking move is simple: **body part** → **job** → **tool idea**. If children can make that chain clearly, they are doing the real work of the standard.



Materials

- Pencil and crayons
- Worksheet pages from this lesson
- Optional: toy animals, animal photos, or a nature book
- Optional: one spoon, one mitten, one umbrella, and one kitchen tongs/grabber for quick comparison examples

👁️ Animal Solution Gallery



Duck Feet

Job: Push water and help the duck swim.

Human idea: Swim fins or flippers that help people move in water.

move through water



Turtle Shell

Job: Protect the turtle's soft body.

Human idea: Helmet, knee pads, or a hard case that protects something fragile.

stay safe



Bird Beak

Job: Pick up, grab, or crack food.

Human idea: Tongs, grabbers, or a picker tool.

pick up and grab



Hedgehog Spines

Job: Help protect the hedgehog from danger.

Human idea: Protective outer covering or gear that keeps someone safer.

protect from harm



Let's Get Started

Step 1 — Notice the Problem Each Animal Solves (8 min)

Look at the Animal Solution Gallery together. For each animal, ask: *"What is hard in this animal's world?"* Then ask: *"What body part helps?"* Keep it concrete: water, danger, reaching food, carrying or moving.

Step 2 — Act It Out (8 min)

Let your child pretend to be each animal. Push hands like duck feet, curl up like a turtle hiding in a shell, pinch fingers like a beak, and freeze like a hedgehog protecting itself. The goal is to feel the job, not just hear about it.

Step 3 — Match Structure to Job (8 min)

Use the worksheet table to match each animal part with what it helps do. If your child gets stuck, return to the sentence frame: *"This part helps it ____."*

Step 4 — Borrow One Good Idea (8 min)

Ask: *"Which animal idea would help a person the most?"* Pick just one. Have your child explain what tool could copy that idea and what the tool would help someone do.

Step 5 — Sketch Tomorrow's Tool Idea (5–8 min)

End by drawing one simple tool idea. Tomorrow's planning lesson will build from this choice, so what matters today is a clear idea, not a perfect drawing.

 **Teaching tip:** If a child starts describing only what the tool looks like, gently pull them back to the job: *"What would it help a person do?"* That is the important engineering move here.



Discussion Questions



Which animal part seemed most helpful?

 Listen for a real job: swim faster, stay safe, pick up food, or protect the body.



What problem does that body part solve?

 Strong answers sound like: *"It helps the animal move in water,"* not just *"It has feet."*



What human tool could copy that idea?

 A strong answer explains both the copied part and the job the tool would do.



Why do engineers study animals and plants?

 *Nature already has lots of smart solutions. Engineers learn from those ideas.*



Parent/Teacher Notes

- **Keep it physical:** This lesson works better when children point, act, and compare—not just listen.
- **Keep the language simple:** “Copy an idea from nature” is better than overloading first graders with technical wording.
- **Do not rush to building yet:** Today's job is to choose a strong nature idea worth copying.
- **What mastery looks like:** A child can say what the body part does and name a sensible human tool idea inspired by it.
- **Helpful correction:** If a child says “The tool should look like a duck,” redirect to function: “What should it help someone do?”



Student Worksheet — Look at Animal Solutions

Name: _____ Date: _____

1 Match the animal part to its job

Animal part	What job does it do?
Duck feet	_____
Turtle shell	_____
Bird beak	_____
Hedgehog spines	_____

2 Choose one good idea from nature

I want to copy the _____ because it helps with
_____.

3 My human tool idea

My tool would help a person _____.

It would copy the animal part by _____.

4 Draw your tool idea

Draw your nature-inspired tool here 



PARENT ANSWER KEY — Detach before giving worksheet to students

Expected matches

- **Duck feet:** help push through water / help the duck swim
- **Turtle shell:** helps protect the turtle
- **Bird beak:** helps pick up or crack food
- **Hedgehog spines:** help protect the hedgehog from danger

Sample strong response

I want to copy a bird beak because it helps pick things up. My tool would help a person grab trash or blocks without using their fingers. It would open and close like a beak.



What success looks like

- Student connects a real animal part to a real job.
- Student chooses a human tool idea that copies the useful feature.
- Student explains the job of the tool, not just how it looks.



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Plan a Nature-Inspired Tool



Engineering Design — Unit 1, Lesson 3 of 6



Grade: 1st Grade



Duration: 30–40 minutes



Subject: Engineering Design



NGSS Standards:

1-LS1-1 — Use materials to design a solution to a human problem by mimicking how plants and/or animals use their external parts to help them survive, grow, and meet their needs.



Learning Objective

Students will **plan** a simple tool inspired by an animal or plant part, **label** how the design will help solve a problem, and name what the tool must do to be helpful.



The Big Idea

Before engineers build, they make a plan. A plan helps us think about what the tool should look like, what materials we might use, and what job it should do. Good plans are simple and clear.



Key vocabulary: plan, label, design, build, mimic, idea



Success rule: A good plan should show what the tool must do—for example, grab, hold, protect, or scoop.



Materials

- Pencil
- Crayons
- Worksheet pages

- Optional sample building materials for discussion



Let's Get Started

Step 1 — Review Yesterday's Idea (5 min)

Talk about the animal part or plant part the child liked best yesterday.

Step 2 — Plan the Tool (10 min)

Ask: "What should your tool do?" and "What shape or part from nature are you copying?" Sentence frame: "My tool will help ___ by using a shape like ___."

Step 3 — Draw and Label (12 min)

Have the child draw the tool and label the important part. Encourage simple words like grab, protect, scoop, slide, hold.

Step 4 — Explain the Plan (8 min)

Have the child explain how the tool will help someone and what part came from nature.



Discussion Questions



Question 1: "What job should your tool do?"

... A strong answer is about the job, like holding, scooping, protecting, or helping someone move something.



Question 2: "What natural part gave you the idea?"

... Students should connect the design to webbed feet, claws, shells, burrs, stems, or another structure.



Question 3: "Why is it helpful to make a plan before building?"

... A plan helps us think first and gives us something to follow when we build.



Parent/Teacher Notes:

- This lesson should feel calm and thoughtful.
- Simple labeled drawings are enough.
- What matters most is the connection between the problem, the natural structure, and the planned solution.



Student Worksheet — Plan a Nature-Inspired Tool

Name: _____ Date: _____

1. What problem will your tool solve?



2. What plant or animal part gave you the idea?



3. Draw and label your tool:

Draw your plan here 

4. My tool will help by:





PARENT/TEACHER ANSWER KEY — Detach before giving worksheet to students

Accept any clear plan that links a natural structure to a simple tool idea and explains the job the tool should do.

Sample plan: *"My tool will help pick up blocks by using a bird-beak shape. The front part closes to grab the block."*



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Build Your First Model



Engineering Design — Unit 1, Lesson 4 of 6



Grade: 1st Grade



Duration: 35–45 minutes



Subject: Engineering Design



NGSS Standards:

K-2-ETS1-2 — Develop a simple sketch, drawing, or physical model to illustrate how the shape of an object helps it function as needed to solve a given problem.



Learning Objective

Students will **build** a first model of their nature-inspired design, **compare** it to their original plan, and explain how it is supposed to help the user.



The Big Idea

A model is a first version. Engineers build models so they can learn what works and what needs to change. A first model does not need to be perfect, but it should still try to do the important job from the plan.



Key vocabulary: model, build, test, compare, shape, function



Design reminder: Before building, ask: *Who is this for, and what must it help them do?*



Materials

- Paper
- Tape
- Scissors
- Craft sticks or cardboard scraps
- Pipe cleaners, string, or other simple classroom materials

- Worksheet pages



Let's Get Started

Step 1 — Review the Plan (5 min)

Look at the drawing from Lesson 3. Ask: "What is the most important part to build first?"

Step 2 — Build the Model (15 min)

Use simple materials to build the first version. Remind students that it is okay if it does not look exactly like the drawing.

Step 3 — Compare Plan and Model (8 min)

Ask: "What stayed the same? What changed? Which change helped your model do its job?"

Step 4 — Get Ready to Test (5 min)

Talk about how they will know if the model helps solve the problem.



Discussion Questions



Question 1: "What part of your model is most important?"

Students should identify the part that does the main job.



Question 2: "How is your model the same as your drawing?"

They can talk about shape, parts, or the job it is supposed to do.



Question 3: "Why is a first model allowed to be imperfect?"

Because engineers learn by building and then improving.



Parent/Teacher Notes:

- Do not over-help. Let the child own the design.
- Messy but thoughtful is fine.
- Today is about turning a plan into something real.



Student Worksheet — Build Your First Model

Name: _____ Date: _____

1. Draw your model after building it:

Draw here 

2. What stayed the same from your plan?

3. What changed?

4. How will you test it?



PARENT/TEACHER ANSWER KEY — Detach before giving worksheet to students

Answers vary. A strong response shows a real connection between the planned design, the built model, and an idea for testing. Example: *"My drawing had a wide scoop to help pick up blocks. My model still has the scoop, but I made the handle shorter so it is easier to hold. I will test whether it can pick up the block without dropping it."*



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Test and Improve Your Design



Engineering Design — Unit 1, Lesson 5 of 6



Grade: 1st Grade



Duration: 30–40 minutes



Subject: Engineering Design



NGSS Standards:

K-2-ETS1-3 — Analyze data from tests of two objects designed to solve the same problem to compare the strengths and weaknesses of how each performs.



Learning Objective

Students will **test** a model, **compare** how version 1 and version 2 perform, and **make one improvement** based on what they noticed.



The Big Idea

Engineers test their ideas. A test gives us information. If a design does not work the first time, that is not failure — it is information that helps us improve.



Key vocabulary: test, improve, strength, weakness, compare, change



Materials

- The model from Lesson 4
- Same building materials used before
- Worksheet pages



Let's Get Started

Step 1 — Decide on a Fair Test (5 min)

Talk about the job of the design and how to check whether it can do that job.

Step 2 — Test the Model (10 min)

Try the design. Watch carefully. Did it help solve the problem? What part worked well? What part needs help?

Step 3 — Change One Thing (8 min)

Choose one improvement. Keep the rest the same so you can tell whether the change helped.

Step 4 — Test Again (8 min)

Run the test again and compare the new result to the first one. Ask: "Did version 2 do the job better, worse, or about the same? What evidence do you have?"



Discussion Questions



Question 1: "What worked well in your first model?"



A strength is something the design already does well.



Question 2: "What needed to change?"



A weakness is something that did not help enough or made the tool harder to use.



Question 3: "Did your change make the design better? How do you know?"



The best answers use evidence from the test.



Parent/Teacher Notes:

- Keep tests simple and visible.
- Students only need one meaningful improvement.
- We are teaching the cycle: build → test → improve.



Student Worksheet — Test and Improve Your Design

Name: _____ Date: _____

Version	What happened?	What changed?	Did it work better?
Version 1			
Version 2			

What was one strength of your design?

What was one weakness?

How do you know version 2 worked better or worse?



PARENT/TEACHER ANSWER KEY — Detach before giving worksheet to students

Answers vary. A strong worksheet shows an actual test, one change, and a comparison between before and after.

- **Sample strength:** “The tool could pick up the object.”
- **Sample weakness:** “The handle bent and made it hard to use.”
- **Sample comparison:** *“Version 2 worked better than version 1 because I made the handle shorter, and it held the object without dropping it.”*



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Share Your Final Design



Engineering Design — Unit 1, Lesson 6 of 6



Grade: 1st Grade



Duration: 30–40 minutes



Subject: Engineering Design



NGSS Standards:

K-2-ETS1-3 — Analyze data from tests of two objects designed to solve the same problem to compare the strengths and weaknesses of how each performs.



Learning Objective

Students will **share** their final design, **explain** how it solves the problem, and **compare** how the final version works better than the first version.



The Big Idea

Engineers share their ideas with other people. Explaining the problem, the design, and the improvement helps others understand how the solution works.



Key vocabulary: final design, explain, improve, share, solve, evidence



Materials

- Final design model
- Worksheet pages
- Optional: family member or partner audience



Let's Get Started

Step 1 — Get Ready to Present (8 min)

Review the original problem, the first idea, and the change that made the design better.

Step 2 — Share the Final Design (10 min)

Have the child show the model and explain what it does, what natural idea inspired it, and what changed after testing.

Step 3 — Ask Questions (8 min)

Listeners can ask: "What problem does it solve?" "What part is most important?" "What did you improve?"

Step 4 — Celebrate the Process (5 min)

Emphasize that the goal was not perfection. The goal was to think like an engineer.

Discussion Questions



Question 1: "What problem does your design solve?"



The problem should be clear and specific.



Question 2: "What part of nature inspired your design?"



Students should name the plant or animal structure they copied.



Question 3: "How did your design get better from the first version to the final version?"



*A strong answer names at least one improvement and what effect it had. Try the sentence frame: "**I changed** ____, **and now my design** ____."*



Parent/Teacher Notes:

- This lesson is about communication and reflection.
- Take photos if you want a record of the child's engineering work.
- The best presentations explain the process, not just the final object.



Student Worksheet — Share Your Final Design

Name: _____ Date: _____

1. My problem was:

2. My design was inspired by:

3. One improvement I made was:

4. I changed _____, and now my design _____.

5. Draw your final design:

Draw here 



PARENT/TEACHER ANSWER KEY — Detach before giving worksheet to students

Answers vary. A complete response identifies the problem, the inspiration from nature, and at least one meaningful improvement.

Sample complete response: *"My problem was picking up blocks. My design was inspired by a bird beak. I made the grabbing part narrower, and now my design can hold the block better than my first version."*



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Engineering Design

GRADE 1 REFERENCE CARD



START WITH A PROBLEM

Name who needs help

Say what is hard

Decide what success should do



LOOK AT NATURE

Animal parts solve problems

Beaks, claws, and stems give ideas

Nature can inspire new tools



PLAN AND BUILD

Draw your tool first

Label the most helpful parts

Build a simple model of version 1



TEST AND IMPROVE

Try the model on the real problem

Notice what worked and what did not

Change one part to make version 2 stronger

Problem → Idea → Plan →

Test what happened

Improve with evidence



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Engineering Design Quiz

Engineering · Unit 1 · Grade 1

Score: 0 / 10 | Questions answered: 0

QUESTION 1

What do engineers do first?

Pick colors

Name the problem

Cut paper

Build the final version

QUESTION 2

Why do engineers look at animals and plants in this unit?

To copy nature's helpful ideas

Because tools must be alive

To avoid testing

Because drawing animals is easier

QUESTION 3

What should a plan show?

Only decorations

The problem, helpful parts, and what the tool should do

Just the child's name

Nothing until testing

QUESTION 4

What is a model?

The biggest final tool only

A smaller version used to try an idea

A random toy

A drawing with no purpose

QUESTION 5

Why do engineers test their design?

To see what worked and what needs improvement

To make it messier

To skip planning

To use more tape

QUESTION 6

What should version 2 do?

Ignore version 1

Change one part to solve a problem better

Use no evidence

Be exactly the same as version 1

QUESTION 7

What makes a final explanation strong?

It only says the tool is cool

It tells the problem, the inspiration, and what improved

It skips the problem

It only names the materials

QUESTION 8

If your tool did not work perfectly the first time, what should you do?

Give up

Hide the model

Use the evidence to improve it

Change the problem

QUESTION 9

What is evidence in engineering?

What you observed during the test

A guess before building

Only what a friend says

The color of the tool

QUESTION 10

Why is nature a good place to get design ideas?

Nature already has structures that help living things survive

Nature never changes

Animals are tools

Nature does not need testing